Pawishrajhen A R

AI Engineer | Machine Learning Engineer
pawishgpt@gmail.com | Chennai, India | LinkedIn | GitHub | Portfolio

Professional Summary

AI Engineer with startup experience shipping Generative AI, computer vision, and LLM systems to production clients. IEEE-published co-author who owns the full path from model development and RAG pipelines to REST APIs, deployment, and client handoff.

Professional Experience

Technical Lead Jan 2025 – Present
KatalyzU — digital studio shipping websites, business software, and AI products *Chennai, India*

- Own technical delivery for client builds across web, mobile, and AI automation on fixed 7- and 30-day timelines.
- Designed Skanda, a multi-tenant billing backend (FastAPI, PostgreSQL) with tenant-scoped ledgers, auth boundaries, and documented REST APIs.
- Delivered The Fuel Depot, a production fuel-delivery platform (React Native field app, admin dashboards, invoicing, live tracking) that replaced paper reconciliation for a Nigerian operator.
- Built and maintain katalyzu.com (Next.js), the studio's public site and client-intake flow.

Freelance AI & Software Engineer 2025 – Present
Independent / Contract (via KatalyzU) *Remote*

- Delivered fixed-scope client projects — marketing sites, admin portals, and internal tools — with full source handoff.
- Built workflow automations and third-party API integrations for founders and small teams.

Featured Projects

SightSpeak | *YOLOv8, LLaMA 3.2, Computer Vision, Text-to-Speech* | GitHub
- Built assistive navigation for visually impaired users: real-time YOLOv8 detection with LLaMA 3.2 spoken guidance.

DermaScan AI | *TensorFlow, OpenCV, LLaMA 3.2/Ollama, Streamlit* | GitHub
- Trained a skin-lesion classifier on HAM10000 (7 classes) with multilingual LLaMA explanations for patients.

MaliciousURLNet | *CNN–BiLSTM, Attention, LLaMA, Cybersecurity* | GitHub
- Built a malicious-URL detector: CNN–BiLSTM encoding, LLaMA threat summaries, and exportable incident reports.

RAG DocChat | *FAISS, Sentence Transformers, Ollama Phi3, Streamlit, OCR* | GitHub
- Built a fully local document Q&A tool with hybrid retrieval, OCR, and source-cited answers — no cloud dependency.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, C
AI / ML: PyTorch, TensorFlow, OpenCV, YOLO, Scikit-learn, XGBoost, Computer Vision, NLP
LLMs & GenAI: Hugging Face Transformers, LLaMA, Ollama, RAG, FAISS, Agentic AI, Model Context Protocol (MCP)
Backend & Frontend: FastAPI, Flask, Node.js, Express, REST APIs, OpenAPI, React, Next.js, React Native, Streamlit
Databases & Cloud: PostgreSQL, MySQL, Firebase, Supabase, Google Cloud Platform, Vercel
DevOps & Tools: Git, Docker, Linux, Jupyter

Publications

- **ReMindAR: Adaptive AR Memory Assistance for Dementia Patients** — co-author; IEEE ICIMA 2026; 82.33% reminder-classifier accuracy. IEEE Xplore
- **Hybrid ARIMA–LSTM Forecasting of Ocean Acidification and Hypoxia** — research lead; IJLRET. DOI

Education

SRM Institute of Science and Technology Chennai, India
B.Tech in Artificial Intelligence; CGPA: 8.44/10 *Jun 2022 – May 2026*

Achievements: RPC 2024 Finalist (Top 12 / 1,000+); HyperVerge Finalist (Top 30 / 200+); Hackheist 2024 Finalist.

Certifications: Machine Learning (Coursera); Oracle OCI Foundations & Data Foundations Associate; Intel Unnati AI + IoT.